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GUARDA

FINALDATASET

Subtítulo (introduzir, se aplicável, ou eliminar o campo)

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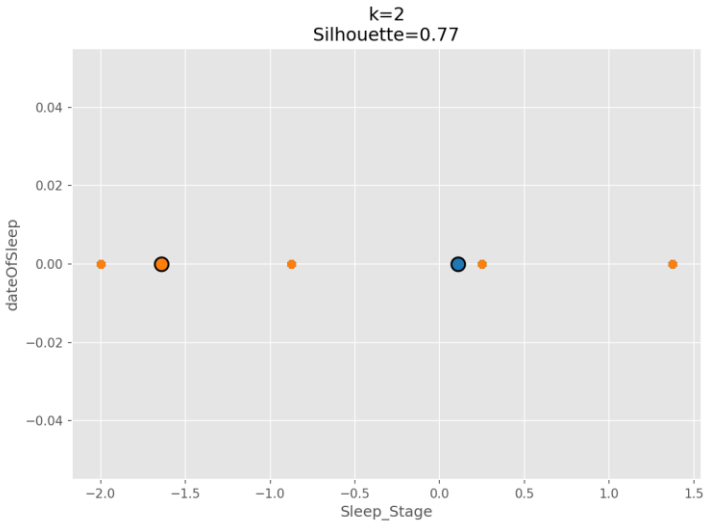
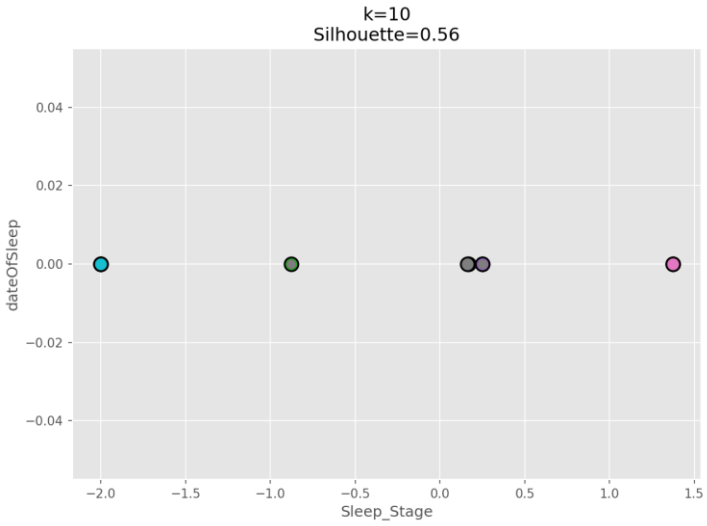
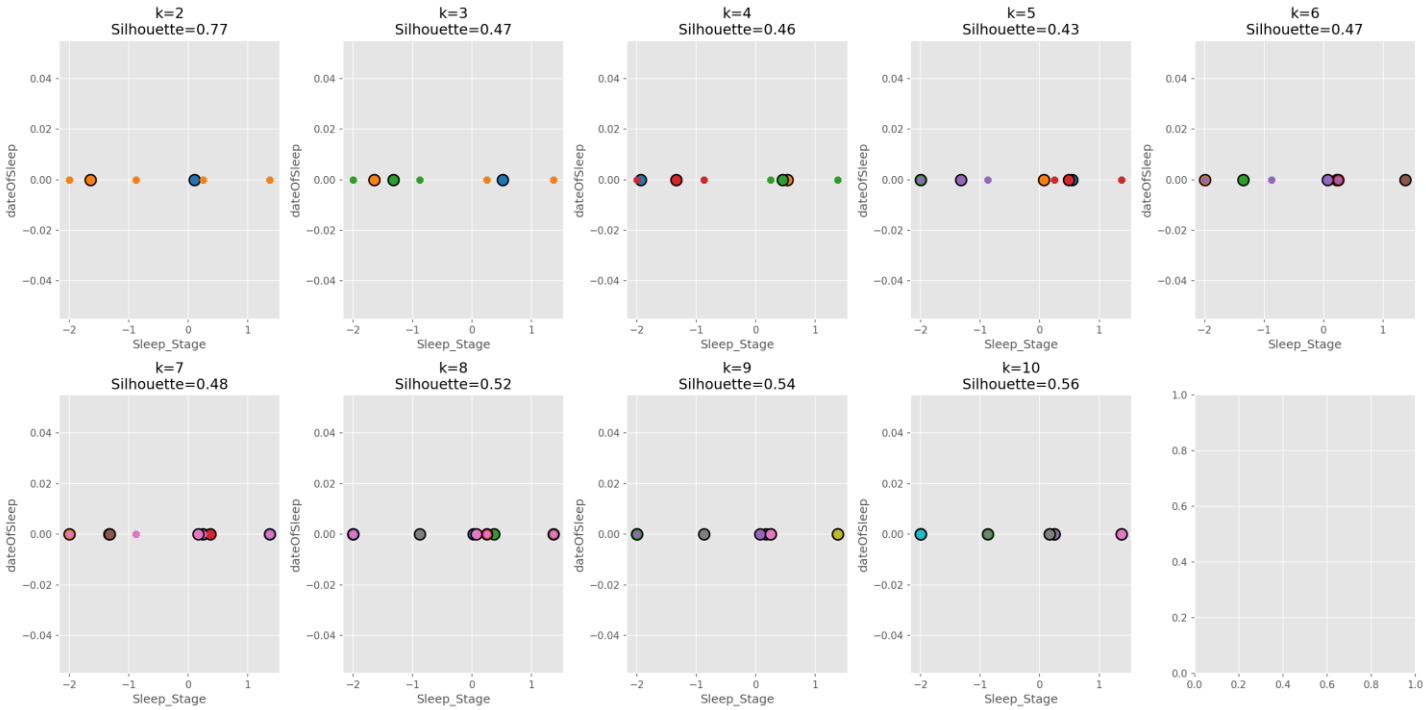
Curso	Clique para introduzir o curso
Unidade curricular	Clique para introduzir a designação da unidade curricular
Ano letivo	Clique para introduzir o ano letivo
Docente(s)	Clique para introduzir o(s) nome(s) do(s) docente(s) da unidade curricular
Data	27/06/2025

The shapes below show the results of a k-means clustering algorithm applied to different datasets, with the number of clusters (k) ranging from 2 to 10. The silhouette score, a measure of how well-separated the clusters are, is provided for each plot.







The visualization presents the results of applying the K-Means algorithm to a dataset consisting of **'Sleep Stage'** and **'Date of Sleep'**. The goal is to identify meaningful clusters that capture patterns in how sleep stages vary over time.
The analysis concludes that the optimal number of clusters is **k = 2**, a conclusion that is clearly supported by both **quantitative evidence** and **visual inspection** of the data.
A visual inspection of the plotted data further confirms this conclusion:

- The data points are organized into **two clearly separated vertical clusters**, resembling **two columns**.
- These columns represent natural groupings in the data—perhaps corresponding to different sleep behaviors or user types on separate nights.
- When **k = 2**, K-Means cleanly separates these two columns into two distinct clusters, **perfectly capturing the existing structure**.

In contrast, when k is increased to 3 or more:

- The algorithm is forced to **arbitrarily divide one or both columns**, despite their internal cohesion.
- This artificial subdivision leads to **overlapping clusters and reduced interpretability**, which is also reflected in the lower silhouette scores.

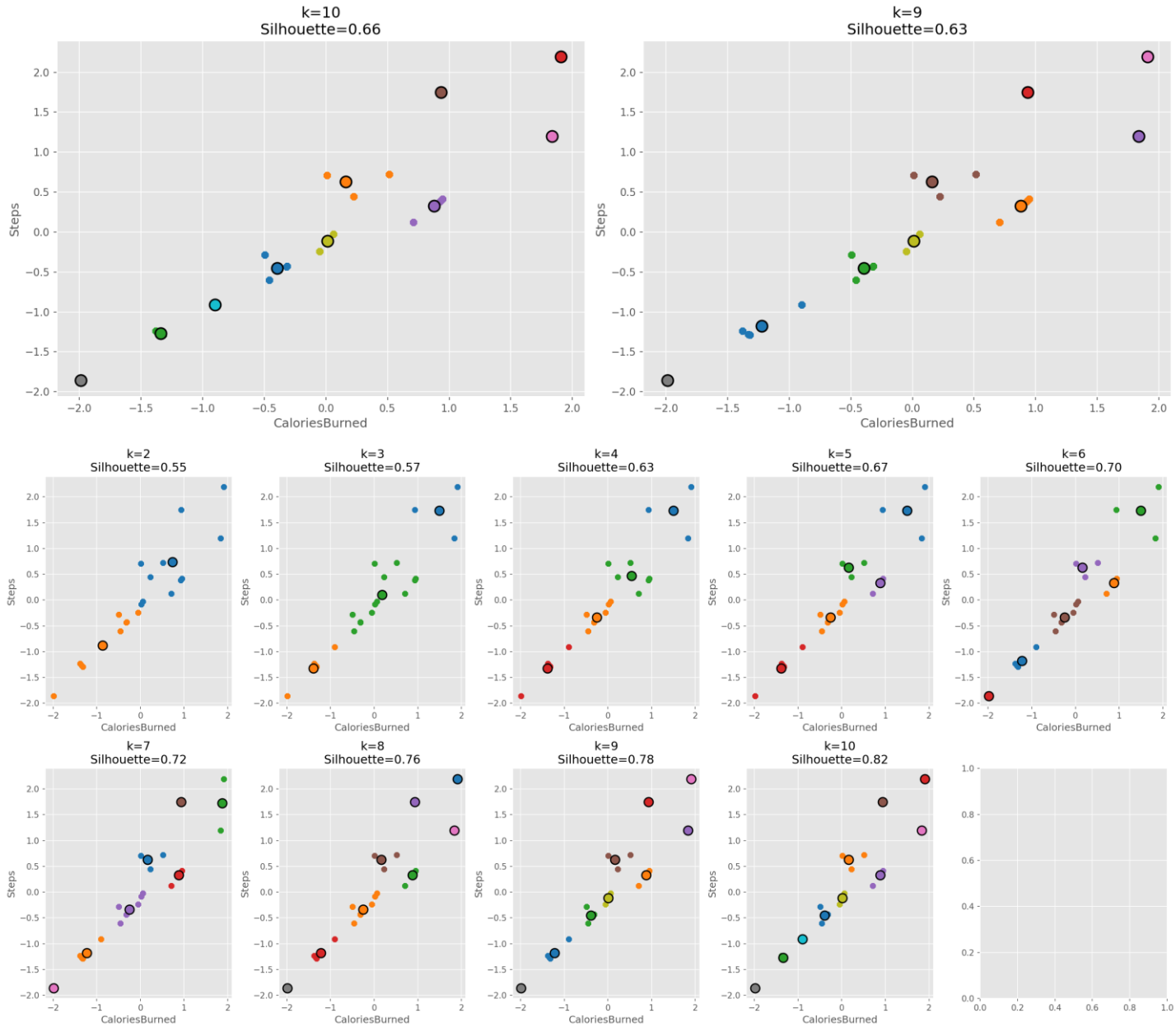
Conclusion:
The structure of this dataset is **inherently binary**, consisting of **two compact and well-separated groups**. Selecting **k = 2** results in:

- The **highest clustering quality** based on Silhouette Score,
- A **clean and interpretable segmentation**,
- And a solution that **aligns with the natural shape of the data**.

Using a higher k would not only reduce cluster performance but also obscure the clarity and meaning of the original data. Therefore, **k = 2 is clearly the optimal choice** for this clustering task.

🔎 Found 6 sleep-related columns:
1. Sleep_Stage
2. dateOfSleep
3. minutesToFallAsleep
4. minutesAsleep
5. timeInBed
6. SleepScore

🏆 Top 2 k-values based on Silhouette Score:
1. k=2, Score=0.767
2. k=10, Score=0.555



The visualization illustrates the result of applying the K-Means clustering algorithm on a dataset with two features: **'Calories Burned'** and **'Steps'**. The algorithm was executed for values of **k ranging from 2 to 10**, in order to determine the most meaningful number of clusters.

The analysis concludes that **k = 10** is the optimal number of clusters, supported by both **quantitative evaluation** (via the Silhouette Score) and **visual inspection** of the resulting cluster structures.
Selecting a lower number of clusters (k < 6) leads to **overgeneralization** and **loss of key information** about user behavior. On the other hand, **k = 10** provides:

- The **highest silhouette score**, indicating superior clustering performance,
- **Visually distinct and interpretable groups**,
- And a segmentation that aligns with the likely real-world diversity of physical activity levels.

Therefore, **k = 10 is the most appropriate and effective choice** for this dataset, enabling detailed, data-driven insights into user behavior based on their steps and calories burned.

🔎 Found 6 relevant columns:
1. CaloriesBurned
2. Steps
3. Distance km
4. stress_1
5. stress_2
6. stress_3

📊 Evaluating silhouette scores for k=2 to 10...
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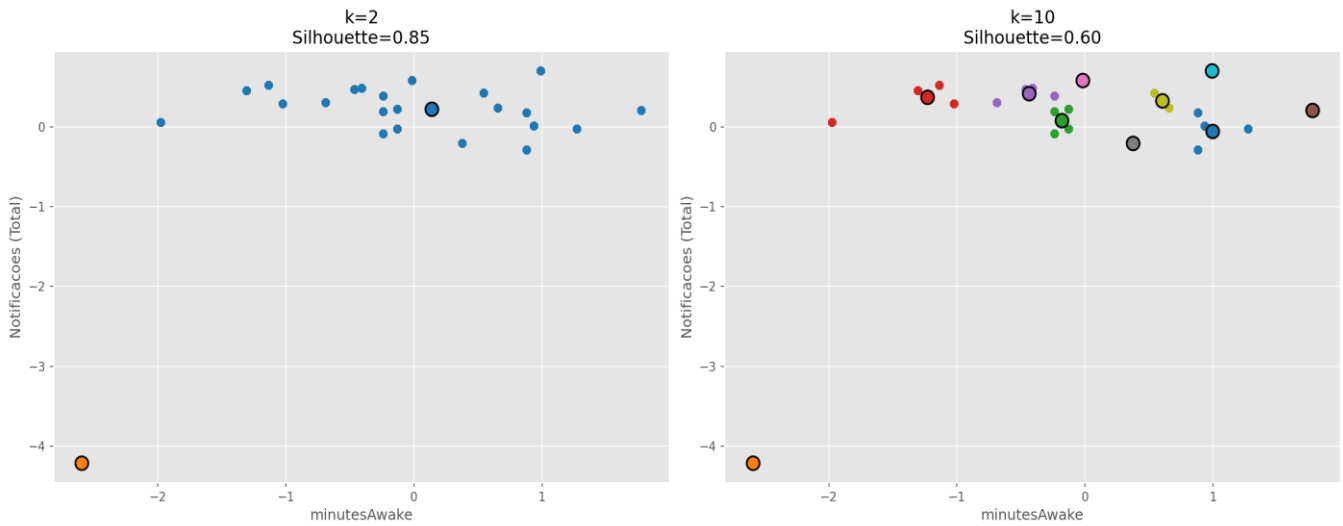
🏆 Top 2 k-values based on Silhouette Score:
1. k=10, Score=0.663
2. k=9, Score=0.634



The visualization presents the outcome of applying the K-Means clustering algorithm with varying values of k (from 2 to 10) on a dataset focused on mobile usage behavior. Specifically, this instance utilizes two features: **'Notificações'** (notifications received) and **'Screen Time'**.
The selection of k = 10 as the optimal number of clusters is based on a combination of quantitative evaluation using the Silhouette Score and visual inspection of the clustering results.
The choice of k = 10 is supported by both the statistical superiority of the Silhouette Score and the visual clarity of the resulting clusters. Lower k values result in oversimplification and loss of important behavioral distinctions, while k = 9 provides an optimal balance between granularity and interpretability.
This cluster configuration offers a robust and insightful segmentation of users based on their device interaction metrics, paving the way for targeted recommendations, personalized settings, or further behavioral analysis
🔎 Found 28 relevant columns:
1. minutesAwake
2. Notificacoes (Total)
3. Abertura de Apps (Total)
4. TempoTela (Total)
5. TempoTela_00
6. TempoTela_01
7. TempoTela_02
8. TempoTela_03
9. TempoTela_04

10. TempoTela_05
11. TempoTela_06
12. TempoTela_07
13. TempoTela_08
14. TempoTela_09
15. TempoTela_10
16. TempoTela_11
17. TempoTela_12
18. TempoTela_13
19. TempoTela_14
20. TempoTela_15
21. TempoTela_16
22. TempoTela_17
23. TempoTela_18
24. TempoTela_19
25. TempoTela_20
26. TempoTela_21
27. TempoTela_22
28. TempoTela_23

📊 Evaluating silhouette scores for k=2 to 10...
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🏆 Top 2 k-values based on Silhouette Score:
1. k=2, Score=0.852
2. k=10, Score=0.600
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